

VCC Workforce Management Platform

Technical Architecture & Development Pipeline

Prepared for: D4 Insight

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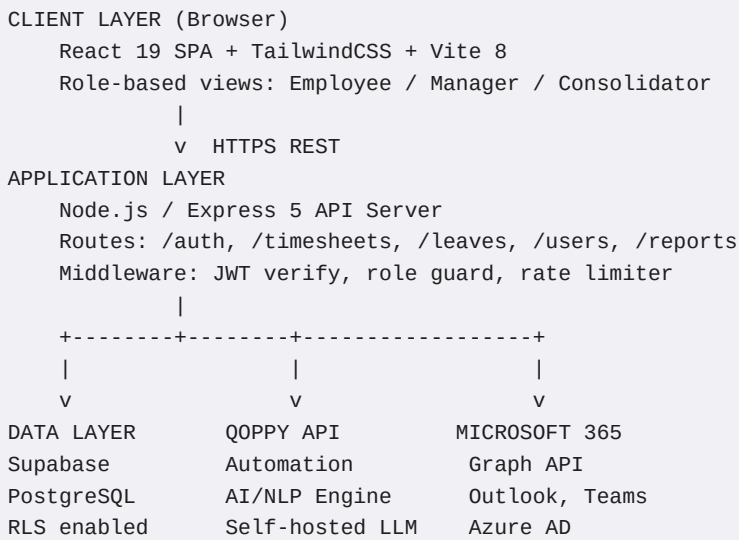
Classification: Confidential

1. System Overview

The VCC Workforce Management Platform is an internal tool for timesheet consolidation, leave management, and resource tracking across D4 Insight's Visual Comfort & Co. engagement. The platform consists of a custom-built web dashboard integrated with Qoppy API for intelligent automation and Microsoft 365 for communication.

The system handles 61+ resources across 9 project groups, automating a fortnightly cycle of timesheet submission, validation, consolidation, review, and client delivery.

2. High-Level Architecture



3. Tech Stack

Category	Technology	Purpose
Frontend	React 19, Vite 8, TailwindCSS 4	Single-page dashboard application
UI Components	Framer Motion, Lucide Icons	Animations, iconography
Backend	Node.js, Express 5	REST API server, routing
Auth	JWT (jsonwebtoken)	Token-based role authentication
Database	PostgreSQL (Supabase)	Cloud DB with Row Level Security
DB Fallback	SQLite (better-sqlite3)	Offline/local development
Automation	Qoppy API	Consolidation, AI review, leave detection
AI/NLP	Qwen LLM (self-hosted)	Email parsing, natural language actions
Communication	Microsoft Graph API	Email send/read, Teams, directory
File Generation	OpenPyXL, xlsx.js	Excel report generation
Hosting	Private VPS (Ubuntu)	Nginx, PM2, systemd services

Version Control	Git	Source code management
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4. Development Pipeline

```
CODE    -->  BUILD    -->  TEST    -->  STAGING    -->  PRODUCTION
|        |        |        |        |
|        |        |        |        |
VSCode   Vite 8    Local Dev  VPS        VPS
Git      ESLint    Mock API   Preview    Live
Branch   Bundle    Qoppy API  Staging DB Prod DB
                Sandbox
```

Branching: feature branch -> review -> main -> staging -> prod

5. Qoppy API Integration

Qoppy provides the intelligent automation layer via REST API. The platform consumes Qoppy endpoints for tasks that require AI processing, scheduled automation, and multi-channel notification orchestration.

API Endpoints Consumed

Endpoint	Function
/qoppy/v1/consolidate	Raw timesheet data -> formatted Excel per project group with summary + per-person breakdown
/qoppy/v1/ai-review	Reviewer email body -> parsed action items (edits, verifications, removals) as JSON
/qoppy/v1/leave-detect	Email content -> detected leave dates, resource name, leave type as structured record
/qoppy/v1/notify	Multi-channel notifications (email + Teams). Handles retries, rate limiting, delivery status
/qoppy/v1/validate	Cross-check timesheet vs leaves, flag anomalies, detect duplicates, enforce business rules

AI Model: Qoppy runs a self-hosted Qwen large language model. No data is sent to OpenAI, Google, or any external AI provider. All inference happens within Qoppy's private infrastructure.

6. Data Flow - Fortnightly Timesheet Cycle

```

Day 1-14: Employees fill timesheets via dashboard
      |
Day 14:  Completion check (platform queries DB)
      Missing? -> Qoppy /notify -> Teams + Email reminder
      |
Day 15:  Consolidator triggers generation
      Platform calls Qoppy /consolidate
      -> 9 project Excel files + 1 master consolidated
      -> Saved to DB + file storage
      |
Day 15:  Send to reviewer via M365 Graph API
      |
Day 15-16: Reviewer replies with feedback (free text)
      Platform detects reply -> Qoppy /ai-review
      -> AI parses: "check with X about date Y"
      -> Qoppy /notify -> verification email + Teams
      |
Day 16-17: Resource replies -> Qoppy /validate -> edits applied
      -> Regenerate -> resend updated file
      |
Day 17:  Approved -> send to clients per project
      Platform sends via M365 Graph API
      |
Day 17+: Finance proceeds with invoicing

```

7. Leave Tracking Pipeline

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Employee sends leave email
  -> CC: leave-tracker@d4insight.com
  |
Platform scans mailbox (Graph API)
  |
Calls Qoppy /leave-detect
  -> parses dates, resource, leave type
  |
Record inserted into Supabase (leaves table)
Dashboard auto-updates in real-time
  |
Leave balance decremented automatically
  |
Threshold check:
  3+ days -> Qoppy /notify -> manager warning email
  5+ days -> Qoppy /notify -> manager + senior escalation
  |
Next timesheet generation:
  Auto-deducts 8h per leave day from resource total
  Reflected in Excel output automatically

```

8. Security Architecture

Layer	Control
Network	HTTPS only (TLS 1.3). No open ports except 443. Nginx reverse proxy.
Authentication	JWT tokens with expiry. Role-based access control (Employee/Manager/Consolidator).
Database	Supabase Row Level Security - users can only query their own data rows.
Qoppy API	API key auth + request signing. Data encrypted in transit (TLS). No data retention by Qoppy.
M365 Integration	OAuth2 client_credentials. Tenant-scoped (D4 only). Customer owns Azure AD app.
AI Processing	Self-hosted Qwen model within Qoppy. No data sent to any external AI service.
File Storage	Excel outputs encrypted at rest (AES-256). Access via authenticated API only.
Audit Logging	All state changes logged: who, when, action, source IP, outcome.
Access Control	SSH key-only server access. Fail2ban. UFW firewall. VPN for admin.
Backups	Automated daily database backups. Encrypted offsite storage.

9. Microsoft 365 Permissions Required

The Azure AD app registration is owned and controlled by D4 Insight. The platform connects using credentials D4 provides. D4 can revoke all access at any time via Azure Portal.

Permission	Type	Purpose
Mail.Send	Application	Send timesheet/notification emails from consolidator account
Mail.Read	Application	Read reviewer replies for AI processing
User.Read.All	Application	Look up employee display names and email addresses
Chat.ReadWrite.All	Application	Send Teams verification messages to resources
Sites.Read.All	Application	Read SharePoint timesheet source data (if applicable)

10. Infrastructure

Environment	Stack	Purpose
Development	Local + SQLite + Vite dev	Feature development, unit testing
Testing	VPS + Supabase test + Qoppy sandbox	Integration testing, email/Teams test mode
Production	VPS + Supabase prod + Qoppy prod	Live deployment for all users

VPS Specification

- OS: Ubuntu 22.04 LTS
- CPU: 4 vCPU
- RAM: 8 GB
- Storage: 100 GB SSD (encrypted)
- Web Server: Nginx (reverse proxy + SSL via Let's Encrypt)
- Process Manager: PM2 (Node.js), systemd (Python workers)
- Firewall: UFW (only port 443 open)
- Security: SSH key-only, Fail2ban, automatic security patches
- Backups: Daily automated to encrypted offsite storage

11. Data Privacy & Controls

Key principles governing data handling within the platform:

1. Data Isolation: Each customer environment is fully isolated. D4 data is never co-mingled with any other organization's data.
2. AI Privacy: The Qoppy AI engine uses a self-hosted Qwen model. No employee data, timesheet data, or email content is ever sent to OpenAI, Google, Microsoft Copilot, or any external AI service.
3. Minimal Access: The platform requests only the minimum M365 permissions needed. No access to SharePoint documents, OneDrive files, or calendar data beyond what is explicitly listed.
4. Customer Control: D4 Insight owns the Azure AD app registration. All credentials are D4-controlled. Access can be fully revoked in under 60 seconds from Azure Portal.
5. No Data Export: Data processed by the platform stays within the platform ecosystem. No data is sold, shared, or used for model training.
6. Retention: Timesheet and leave data retained per D4's policy. On contract termination, all data is permanently deleted within 30 days.